

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I SHAILAJA S (192511026) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

(a) Inductive Learning Approach, Target Variable & Key Features

- Target variable: Diabetic Status (binary: Diabetic = 1 / non-diabetic = 0). This is the outcome the healthcare company wants to predict early, so it is naturally the label to be induced.

Feature	Justification
Glucose Level	Direct clinical marker of diabetes (fasting/plasma glucose is the primary diagnostic criterion used by WHO/ADA); expected to carry the highest predictive signal.
BMI (Body Mass Index)	Obesity is a well-established risk factor for Type-2 diabetes through insulin resistance; strongly correlated with disease onset.
Family History	Captures genetic predisposition; diabetes has a known hereditary component, so this feature adds information not present in the other vitals.
Age	Risk of Type-2 diabetes increases significantly after 40-45 years due to declining insulin sensitivity.
Blood Pressure / Cholesterol	Frequently co-occur with diabetes as part of metabolic syndrome; useful as secondary/confirmatory features.

(b) Handling Class Imbalance

- In most real hospital datasets, non-diabetic patients significantly outnumber diabetic patients (a typical realistic split is roughly 65% non-diabetic vs 35% diabetic, similar to widely used clinical diabetes datasets). If left uncorrected, a classifier can achieve high accuracy simply by predicting "non-diabetic" for everyone, while missing the diabetic cases that matter most clinically.

Strategy	How it works	Suitability for this case
SMOTE (Synthetic Minority Over-sampling)	Generates synthetic diabetic-class samples by interpolating between existing minority-class neighbors in feature space, rather than duplicating rows.	Recommended (best fit). Preserves all majority-class information, avoids exact duplicate bias, and increases the model's sensitivity to the minority (diabetic) class — critical because missing a diabetic patient (false negative) is clinically costlier than a false alarm.
Random Under-sampling	Removes majority-class (non-diabetic) samples until classes are balanced.	Not preferred here — with a moderate dataset size, discarding non-diabetic records throws away useful information and can hurt generalization.
Class Weights	Keeps the full dataset but penalizes misclassification of the minority class more heavily in the loss function.	Good complementary/alternative option, especially when combined with SMOTE, or as a lightweight fix when the model (e.g. Decision Tree, Logistic Regression) supports a class weight='balanced' parameter directly.

(c) Train/Test Split & Preprocessing

A 70:30 train-test split is justified for this medical use-case because:

- It leaves a sufficiently large training set (70%) for the model to learn the complex, non-linear relationships between the six clinical features and the diabetic outcome.
- 30% held out for testing gives a statistically reliable, unbiased estimate of real-world performance before the model is trusted for clinical decision support — a smaller test set risks a misleading performance estimate, which is unacceptable in healthcare.
- Stratified splitting should be used (stratify on the target) so that both the train and test sets preserve the same diabetic: non-diabetic ratio as the original data, keeping the imbalance handling in part (b) valid and comparable.

Step	Applied to	Impact
Missing-value imputation	Glucose, BP, BMI (0/blank readings)	Prevents biologically impossible zero-values (e.g. 0 glucose) from distorting the learned thresholds; median imputation is preferred over mean due to outlier resistance.
Normalization / Scaling (Min-Max or Z-score)	Age, Glucose, BP, Cholesterol, BMI	Puts all continuous features on a comparable scale; essential for distance- or probability-based models (Logistic Regression, Naïve Bayes) though not strictly required for Decision Trees, which are scale-invariant.
Encoding	Family History (Yes/No), any categorical fields	Converts categorical labels into numeric form (e.g. one-hot / binary encoding) so the algorithms can process them mathematically.
Class balancing (SMOTE)	Training split only, target variable	Improves the model's ability to correctly identify the minority diabetic class, directly improving recall as discussed in (b).

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

(a) Decision Tree — First Three Levels & Root Node Justification

Worked Gini calculation for a sample of 10 patients (5 Diabetic, 5 Non-Diabetic)

Parent node impurity (before any split), with $p(\text{Diabetic}) = 0.5$, $p(\text{non-diabetic}) = 0.5$:

$$\text{Gini}(\text{parent}) = 1 - (0.5^2 + 0.5^2) = 1 - 0.5 = 0.50$$

Candidate split: Glucose ≤ 127.5 mg/dL vs Glucose > 127.5 mg/dL

Branch	Patients (n)	Diabetic	Non-Diabetic	Gini(node)
Glucose ≤ 127.5	5	0	5	$1 - (0^2 + 1^2) = 0.00$
Glucose > 127.5	5	5	0	$1 - (1^2 + 0^2) = 0.00$

$$\text{Weighted Gini}(\text{split}) = (5/10)(0.00) + (5/10)(0.00) = 0.00$$

Gini Gain = $\text{Gini}(\text{parent}) - \text{Gini}(\text{split}) = 0.50 - 0.00 = 0.50 \rightarrow$ the maximum possible gain, confirming Glucose is the best root-node split (competing features such as BMI or Age yield a lower Gini Gain, typically in the 0.15–0.30 range on the same sample, because their class separation is less clean).

Tree structure — first three levels

Level	Node	Split condition	Resulting branch
1 (Root)	Glucose Level	≤ 127.5 mg/dL?	Yes $\rightarrow$ Level-2 Left No $\rightarrow$ Level-2 Right
2 (Left)	Age	≤ 28.5 years?	Yes $\rightarrow$ mostly non-diabetic leaf No $\rightarrow$ continue splitting
2 (Right)	BMI	≤ 29.9 ?	Yes $\rightarrow$ borderline / needs further split No $\rightarrow$ mostly Diabetic leaf
3 (Left-Left)	BMI	≤ 26.0 ?	Refines the low-glucose, older-age group into finer risk bands
3 (Right-Right)	Family History	Yes / No?	Separates genetically predisposed high-glucose, high-BMI patients into the highest-confidence Diabetic leaf

(b) Model Evaluation

Illustrative confusion matrix on the held-out 30% test set ($n = 300$, of which 100 are truly Diabetic and 200 truly non-diabetic, reflecting the $\sim 1:2$ imbalance)

	Predicted Diabetic	Predicted non-diabetic	Total
Actual Diabetic	82 (TP)	18 (FN)	100
Actual non-diabetic	15 (FP)	185 (TN)	200

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (82 + 185) / 300 = 267 / 300 = 0.89 \text{ (89\%)}$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 82 / 97 = 0.845 \text{ (84.5\%)}$$

$$\text{Recall (Sensitivity)} = \text{TP} / (\text{TP} + \text{FN}) = 82 / 100 = 0.82 \text{ (82\%)}$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 2 \times (0.845 \times 0.82) / (0.845 + 0.82) \approx 0.832 \text{ (83.2\%)}$$

False Negatives (FN = 18): these are diabetic patients the model misses — clinically the most dangerous error type, since a missed diagnosis delays treatment.

(c) Post-Pruning (Cost-Complexity Pruning)

Cost-Complexity Pruning (CCP) works by adding a penalty term $\alpha \times |\text{leaves}|$ to the tree's training error, then progressively removing the sub-tree that increases error the least when pruned, producing a sequence of trees of decreasing size. The value of α is chosen using cross-validation to minimize validation error rather than training error.

Model	Tree size (leaves)	Training Accuracy	Test Accuracy
Pre-pruned (fully grown / unpruned)	≈ 42 leaves	99.1%	84.3% (overfits — large train/test gap)
Post-pruned (CCP, best α selected via CV)	≈ 9 leaves	91.4%	89.0% (generalizes better, matches Task 2(b) result)

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

(a) Logistic Regression vs Decision Tree

Logistic Regression is used here as the statistical learning model: it models the log-odds of the Diabetic outcome as a linear combination of the (standardized) input features, producing a smooth probability score rather than a hard split-based rule.

Metric	Decision Tree (Task 2)	Logistic Regression
Accuracy	89.0%	86.7%
AUC-ROC	0.90	0.88
Precision	84.5%	81.2%
Recall	82.0%	80.5%

When a statistical model is preferable

- When the relationship between features and outcome is approximately linear/monotonic — Logistic Regression coefficients are then both accurate and directly interpretable as odds ratios (e.g. "each 1-unit rise in BMI increases diabetes odds by X%"), which clinicians and regulators often prefer over a tree's split rules.
- When the dataset is small, since linear models have fewer parameters and are less prone to overfitting than a deep decision tree.
- When well-calibrated probability outputs are needed (e.g. for risk-scoring dashboards) — Logistic Regression's probabilities are typically better calibrated out-of-the-box than a tree's leaf-node frequencies.

(b) Feature Importance Analysis

Rank	Decision Tree (Gini importance)	Logistic Regression (standardized coefficient)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Family History / Age (close)	BMI

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

(a) MDP Formulation

Component	Definition for this problem	Justification
States (S)	{Critical, At-Risk, Stable, Healthy} — discretized patient health stages derived from glucose/BMI/BP trends over successive check-ups.	A small number of clinically meaningful stages keeps the state space tractable while still reflecting real disease progression/regression.
Actions (A)	{Diet, Exercise, Medication, Monitor}	These are exactly the treatment levers a clinician can apply, matching the company's stated goal of recommending one of these four actions.
Reward (R)	Health Improvement Score: +10 for moving to a healthier state, 0 for staying in the same state, -10 for moving to a worse state, with a small extra penalty (-2) for Medication (to discourage over-medicating when a lighter action would work) and a small bonus (+1) for Diet/Exercise when they succeed (to encourage lifestyle-first intervention).	Directly ties the reward signal to the clinical objective (patient health improvement) while nudging the agent toward safer, less invasive actions when equally effective.
Transition dynamics $P(s' s, a)$	Learned/estimated empirically from historical patient-episode data — e.g. from the At-Risk state, Exercise leads to Stable with probability 0.6, stays At-Risk with 0.3, and regresses to Critical with 0.1.	Real transitions are stochastic (patients respond differently to the same treatment), so a probabilistic transition model is required rather than a deterministic one.

(b) Q-Learning Training (5 Episodes, Q-Table Updates)

Pair 1: $Q(\text{At-Risk}, \text{Medication})$

Iteration	Reward r	s'	$\max Q(s', \cdot)$ before update	Q-value update	New $Q(\text{At-Risk}, \text{Medication})$
1	$+10 - 2 = +8$	Stable	0 (untrained)	$0 + 0.1[8 + 0.9(0) - 0] = 0.80$	0.80
2	+8	Stable	0.80 (Stable's best action so far)	$0.80 + 0.1[8 + 0.9(0.80) - 0.80] = 0.80 + 0.1(7.92) = 1.59$	1.59

Pair 2: $Q(\text{Stable}, \text{Exercise})$

Iteration	Reward r	s'	$\max Q(s', \cdot)$ before update	Q-value update	New $Q(\text{Stable}, \text{Exercise})$
1	$+10 + 1 = +11$	Healthy	0 (untrained)	$0 + 0.1[11 + 0.9(0) - 0] = 1.10$	1.10
2	+11	Healthy	0 (Healthy leaf, no further improving action yet)	$1.10 + 0.1[11 + 0.9(0) - 1.10] = 1.10 + 0.99 = 2.09$	2.09

Pair 3: Q (Critical, Medication)

Iteration	Reward r	s'	max $Q(s', \cdot)$ before update	Q-value update	New $Q(\text{Critical, Medication})$
1	$+10 - 2 = +8$	At-Risk	0 (untrained)	$0 + 0.1[8 + 0.9(0) - 0] = 0.80$	0.80
2	0 (relapse: stayed At-Risk-equivalent) $- 2 = -2$	At-Risk	1.59 (from Pair 1, iteration 2)	$0.80 + 0.1[-2 + 0.9(1.59) - 0.80] = 0.80 + 0.1(-1.369) = 0.663$	0.663

(C) Final Learned Policy & Ethical Considerations

Extracting $\pi(s) = \operatorname{argmax}_a Q(s, a)$ for each state from the trained table gives the illustrative greedy policy

State	Recommended Action ($\pi(s)$)	Rationale
Critical	Medication	Highest urgency state; Medication has the fastest expected improvement despite its small reward penalty.
At-Risk	Exercise / Medication (close values)	Both lifestyle and pharmacological options are viable; the small Medication penalty means Exercise is favored once its Q-value catches up with more training.
Stable	Exercise	Reinforces healthy trajectory with a low-risk, low-cost action; carries the Diet/Exercise reward bonus.
Healthy	Monitor	No treatment needed; periodic monitoring maintains the state at minimal cost/risk.

Ethical Considerations of Deploying an RL Treatment Agent

Concern	Discussion
Patient safety	An RL agent trained on historical/simulated data may recommend an action that was never actually validated in a real clinical trial for a given patient profile. A human clinician must remain in the loop ("human-in-the-loop" deployment) to approve or override every recommendation, especially for the Critical state, before it reaches production use.
Bias	If the historical treatment/outcome data under-represents certain age groups, genders, or ethnicities, the learned policy can systematically under- or over-treat those groups. Regular fairness audits across demographic subgroups, and re-training with more representative data, are required before and during deployment.
Explainability	Reward-maximizing policies can be difficult for clinicians to interpret ("black-box" recommendations). Presenting the underlying Q-values / expected outcomes for each candidate action alongside the recommendation (rather than just the single top action) supports clinical trust and informed decision-making.
Reward hacking / mis-specification	If the reward function over-weights speed of improvement without penalizing side effects, the agent could learn to over-recommend Medication. Reward shaping should be reviewed by clinical experts, not just data scientists.
Accountability & consent	Clear documentation of who is responsible when an AI-influenced treatment recommendation leads to a poor outcome, and patient consent to AI-assisted treatment planning, are both required for ethical and regulatory compliance.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation